REFERRING SERVICE

Service	**NeuroReach Rehabilitation Unit**
Department	[redacted]
Clinician	[redacted]
Supervising	[redacted]

PATIENT IDENTIFIERS

Patient name	**Krishna B.**
Patient ID	**NR-0042**
Date of birth	**1958-03-12**
Affected side	**Left (right hemisphere)**
Session	**Week 4 · #11 · 14:20**

Session context

Blended-mode (Mode B) **Star Catcher — Reach & Collect**, 40-minute clinic slot. Fusion weighting w_eeg 0.60 / w_kin 0.40. Abbreviated daily calibration refresh (per-session model v11). Line frequency 60 Hz. Rest-pose captured at session start for relative kinematics.

Session summary

Patient completed a full 40-minute Blended-mode session (22 min active gameplay; 58 of 62 trials valid). EEG sensorimotor engagement was present and correctly lateralised to the affected right hemisphere (mean mu-ERD −22% at C4; laterality index −0.29). Movement range is improving week-on-week (shoulder flexion 78°, 87% of target). One fatigue flag fired at 34 min. Overall engagement good; **wrist extension** and **trunk compensation** flagged for focus next session.

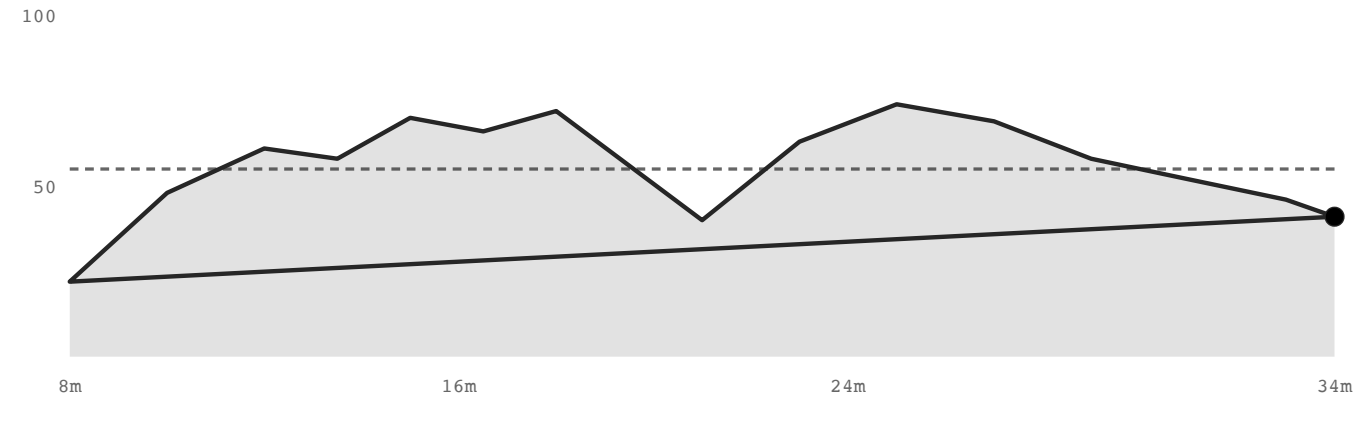
Session metrics

weighting w_eeg 0.60 · w_kin 0.40

MEASURE	RESULT	REFERENCE / TARGET	INTERPRETATION
Active gameplay	22:10 min	2 × ~11 min blocks	—
Setup time	7.2 min	< 10 min	✓ WITHIN TARGET
Trials valid / attempted	58 / 62	—	4 excluded (artifact)
Stars collected	47	reach targets	targets met
Mean IntentScore	64 / 100	in-session norm	peak 89
Difficulty level	6	adaptive staircase	↑ +1 VS LAST WEEK
Success rate	71%	70–80% flow band	✓ IN BAND

IntentScore across the session

0–100 · mean 64 · smoothed 20 Hz



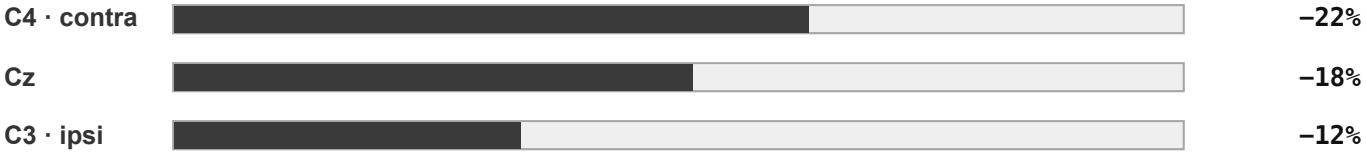
EEG — sensorimotor rhythm

mu 8–12 Hz · C3/Cz/C4 · N = 58 valid trials

MEASURE	RESULT	REFERENCE	INTERPRETATION
Mu-ERD C4 (contralateral)	−22% mean · −31% peak	ERD > 15% at C4	✓ PRESENT
Mu-ERD C3 (ipsilateral)	−12%	—	lower, as expected
Mu-ERD Cz (central)	−18%	—	bilateral
Laterality index (LI)	−0.29	toward affected hemi	✓ LATERALISED
Beta rebound (ERS)	+24%	post-window present	✓ PRESENT
ERD onset latency	0.8 s	cue → threshold	prompt
Trial-to-trial CV	0.34	lower = better	▲ MODERATE

Mean mu-ERD by electrode

longer bar = stronger desynchronization · affected = R hemi (C4)



Decode quality

CSP + LDA · per-session model v11

MEASURE	RESULT	REFERENCE	INTERPRETATION
Calibration CV accuracy	68%	≥ 60% to run EEG-led	✓ EEG-LED
Classifier confidence	0.71 mean	distance-to-hyperplane	—
Time classifier-reliable	74%	—	26% band-power fallback

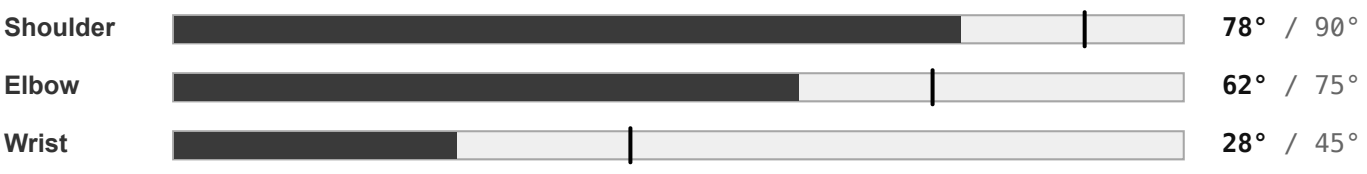
Movement & range of motion

MediaPipe Pose · Tier 2 — relative, un-goniometer-validated

MEASURE	RESULT	REFERENCE / TARGET	INTERPRETATION
Shoulder flexion (peak)	78°	90° target	87% · TIER 2
Elbow extension (peak)	62°	—	+6° vs wk 3 · TIER 2
Wrist extension (peak)	28°	45° target	▲ LIMITED · TIER 2
ROM vs target (avg)	82%	across 58 reaches	—
Movement units / reach	3.2	1–2 healthy	▲ FRAGMENTED
Movement onset latency	1.1 s	cue → reach	—

Peak range of motion vs target

degrees · Tier 2 — relative, pending goniometer validation



Achieved (peak) | Session target

Compensation & data quality

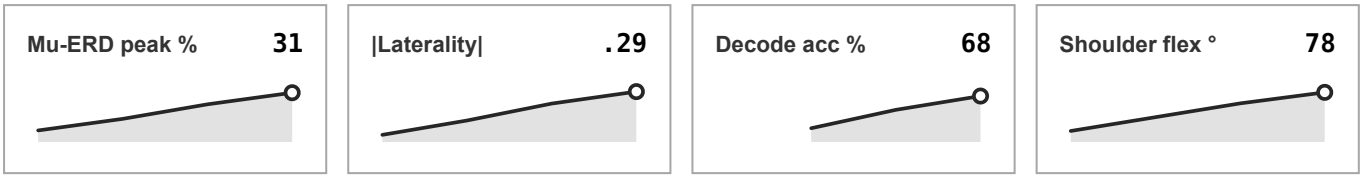
clinical guardrails

MEASURE	RESULT	REFERENCE	INTERPRETATION
Trunk-lean flags	18% of reaches	keep minimal	▲ WATCH · mean 4.2 cm
Signal-quality uptime	91%	—	✓ GOOD · artifact 9%
Camera-link uptime	98%	—	✓ STABLE · 1× EEG-only 6 s
Engagement endurance	14.5 min	above threshold	66% gameplay · bout 95 s
Fatigue proxy	fired 34:00	—	■ FLAG · consider ending

Action — Fatigue flag fired at 34 min (combined decline in signal quality, classifier confidence, and kinematic engagement during block 2). Session wound down shortly after. Consider a shorter block 2 or an earlier rest next session.

Four-week trend

un-normalized proxies · vs FMA-UE where available



METRIC	WK 1	WK 2	WK 3	WK 4	TRAJECTORY
Mu-ERD C4 (peak %)	18	22	27	31	
Laterality index (|LI|)	0.11	0.17	0.24	0.29	
Decode accuracy (%)	—	61	65	68	
Shoulder flexion (peak °)	64	69	74	78	
Difficulty level	3	4	5	6	
FMA-UE (clinician-entered)	—	31	—	35	

Concordance — System proxies (ERD, LI, decode accuracy, ROM) are trending up in step with the clinician-entered FMA-UE (31 → 35 over the block). Continue periodic formal assessment to anchor the trend.

Signals & features analysed

NeuroReach MI + ROM panel v2

Montage — g.tec Unicorn Hybrid Black, 8 ch @ 250 Hz, mastoid reference (M1/M2)

Electrodes: Fz, C3, Cz, C4, Pz, PO7, Oz, PO8

Features (24): mu-band power [C3, Cz, C4], beta-band power [C3, Cz, C4], ERD% per channel, beta rebound / ERS, laterality index, CSP+LDA decode, classifier confidence, fused IntentScore, amplitude gate, IMU movement gate, Fz EOG-proxy, shoulder flexion, elbow extension, wrist extension, ROM-vs-target, movement units / jerk, movement onset latency, trunk displacement, landmark visibility, engagement endurance, artifact rate, signal-quality uptime, camera-link health, adaptive difficulty level

Coverage — 24 features · N = 58 valid trials · 250 Hz acquisition → 20 Hz score · classifier reliable 74% of session

Notes & reliability

Reliability tiers. Tier 1 (validated pipeline — report as-is): ERD, laterality, decode accuracy, IntentScore, engagement, difficulty. Tier 2 (relative & trend only, pending goniometer validation): all ROM / joint-angle values.

N & denominators. EEG means over 58 valid trials (4 of 62 excluded for artifact). ROM over 58 completed reaches. Percentages are of active gameplay unless noted.

Normalization. The 0–100 IntentScore is per-session adaptive-normalized and is not comparable across days — the 4-week trend uses un-normalized ERD%, angles, and accuracy.

Laterality. Left-hand task → contralateral right hemisphere (C4). Negative LI indicates correct lateralization toward the affected hemisphere.

Reported by (Occupational Therapist)	[redacted]	[redacted]
Authorised by (Consultant)	[redacted]	Date:

NeuroReach is an engagement and biofeedback tool used as a clinician-supervised adjunct. Metrics describe tracked engagement and movement over time; they are not a diagnostic or a standalone measure of recovery. Not for autonomous clinical decision-making.

Great session, Krishna! ⭐

Star Catcher · 10 July 2026 · about 40 minutes

🏆 **New personal best — Level 6!**

⭐

47

Stars caught

reaching for each one

⏱️

15 min

In the zone

focused & engaged

🎯

Level 6

Reached today

up from Level 5 last week

💪

82%

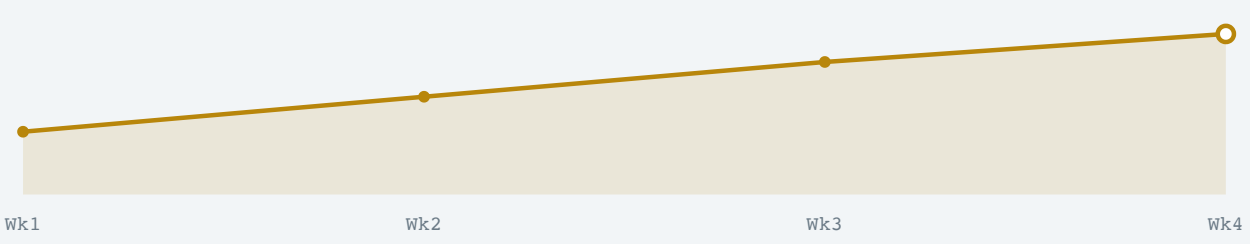
Reach goal

of today's target distance

Reach progress 82%

Your reach, week by week

a little farther each week 🌟



You reached a little **farther and higher** than last week and stayed focused right through both rounds. Nice work keeping your **shoulders steady** while you reached. Next time we'll practice a few more **wrist stretches** toward the stars. 🌟

Shared by your therapist · Keep it up! · No medical data leaves the clinic.